

Proxmox Virtual Environment

Completely Open Source Hypervisor

Jamilur Rahman
Manager (Systems & Cloud)
Bangladesh Research and Education Network (BdREN)



www.linkedin.com/in/jamilur-rahman1/

Course *Roadmap*

Open-Source Enterprise Backup Solution - PBS

Disaster Recovery: DC-DR Concept

Proxmox VE Firewall

Cloud Platform Migration

High Availability (HA) - PVE HA Manager

Deploy Hyper-Converged Ceph Cluster

Cluster Manager

IAM – User Management

Host System Administration

Single Proxmox node setup

Proxmox VE: Concept, Architecture, Technologies

Cloud Computing & Server Virtualization

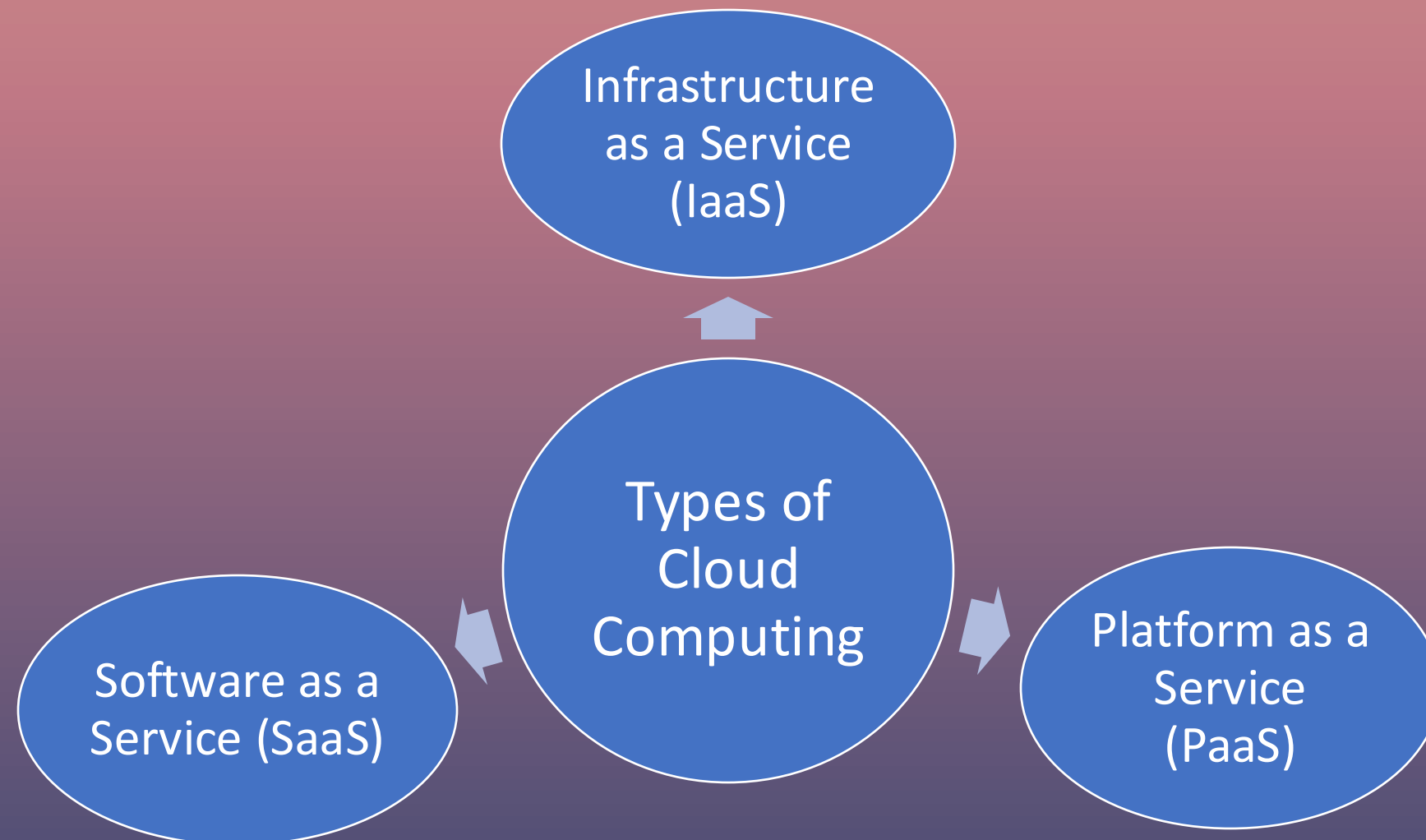


Intended Audience

- System Administrators/ Engineers
- Network Engineer
- IT specialist
- Anyone who wants to deploy virtualization on Home-Lab

Cloud Computing

- Cloud Computing is an on-demand delivery of IT resources.
- Virtualization is the core component of Cloud computing

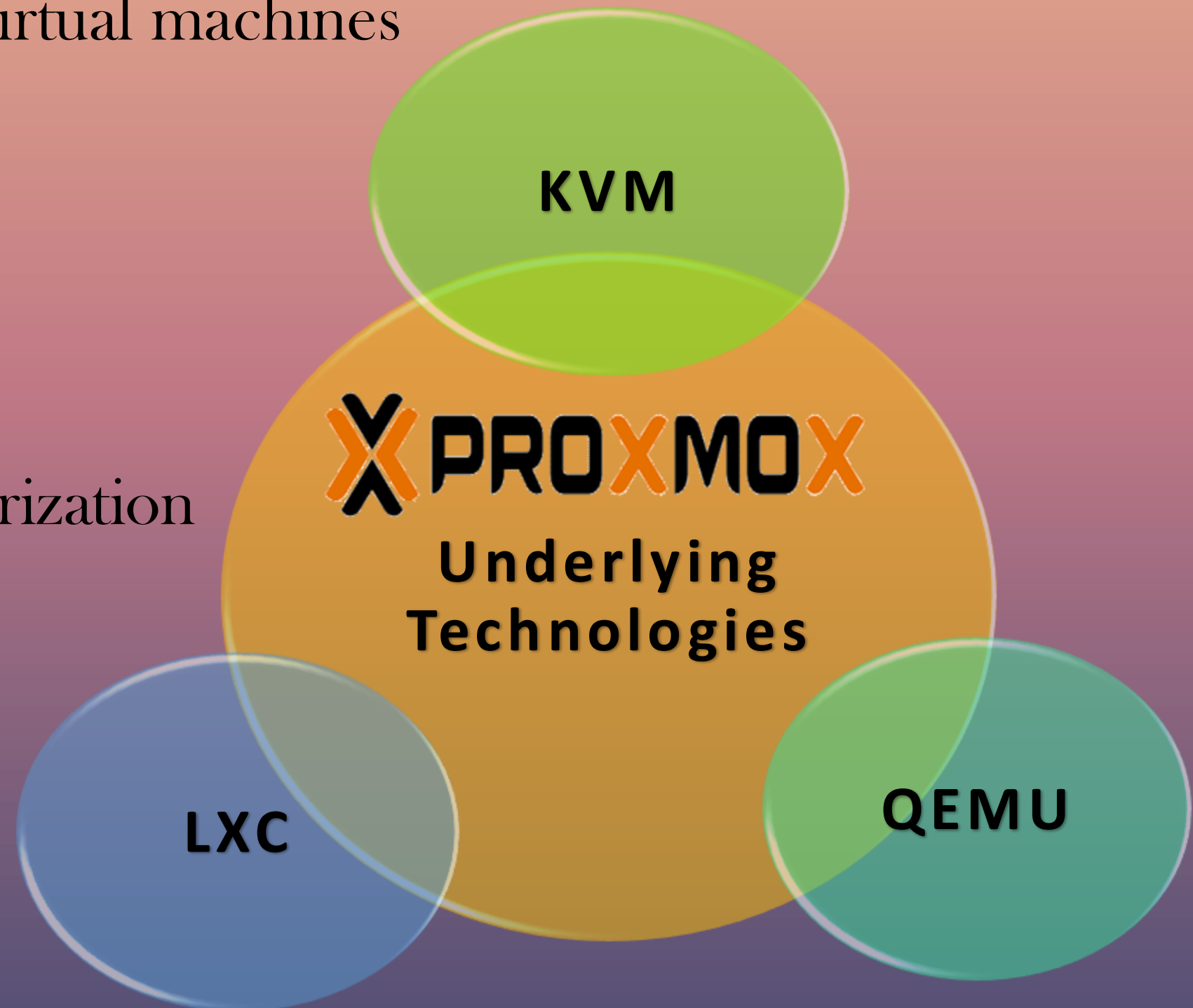


Existing Virtualization Software (Hypervisor) in the Market

- VMware ESXi
- Microsoft Hyper-V
- OpenStack
- Proxmox VE
- Nutanix AHV
- Xen / Citrix XenServer
- And So on....

Overview of Proxmox VE

- ✓ Proxmox VE is a platform to run virtual machines and containers.
- ✓ It is based on Debian Linux
- ✓ Completely Open Source Code
- ✓ Offers Virtualization and Containerization
- ✓ Type-2 Hypervisor



Introduction on KVM

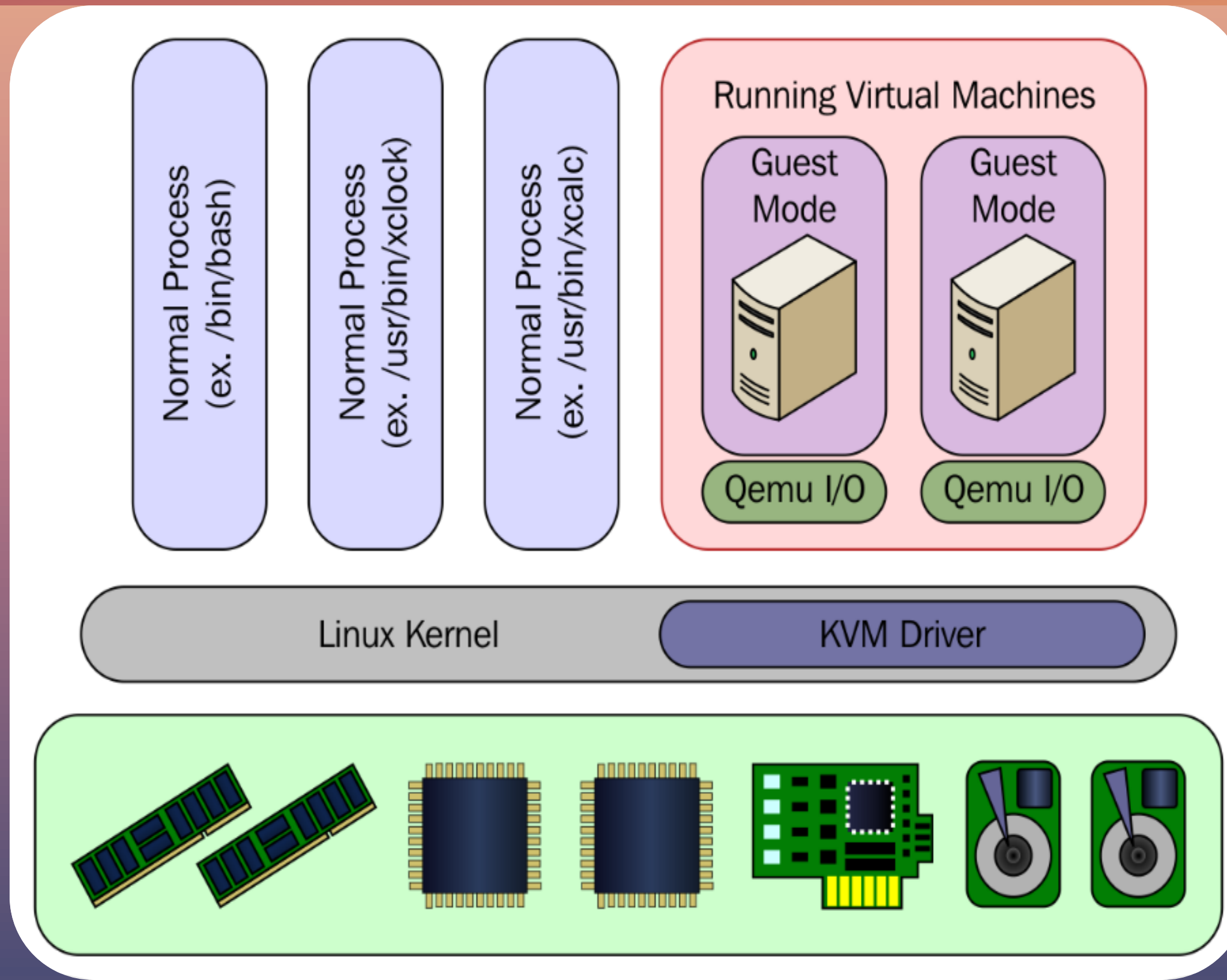
Kernel-based Virtual Machine

- ✓ Leading Virtualization Technology for Linux
- ✓ Works on top of Operating System (Linux Kernel)
- ✓ Used with QEMU, called QEMU-KVM
- ✓ Allows the Kernel to function as a Hypervisor
- ✓ Open Source Technology
- ✓ Type-2 hypervisor



Architecture of KVM

Kernel-based Virtual Machine



Features and Use Case of KVM

Kernel-based Virtual Machine

FEATURES

- 1 Paravirtualization (VirtIO)
- 2 Hot Plug vCPU
- 3 Dynamic Memory Management
- 4 Live Migration

*Using KVM technology,
you can develop your
own Virtualization
Platform.*

Use Case

Why KVM & QEMU works together

- ✓ **KVM** provides **CPU virtualization** in the Linux kernel; **QEMU** provides **everything else** a VM needs
- ✓ **QEMU (Quick EMUlator)**
 - It provides **device emulation** (disk, NIC, GPU, serial, chipset, firmware), VM lifecycle (start/stop), **live migration**, snapshots, SPICE/VNC display
- ✓ **KVM without QEMU:** You'd have CPU virtualization but **no** device model, no management tooling—so not practical by itself.

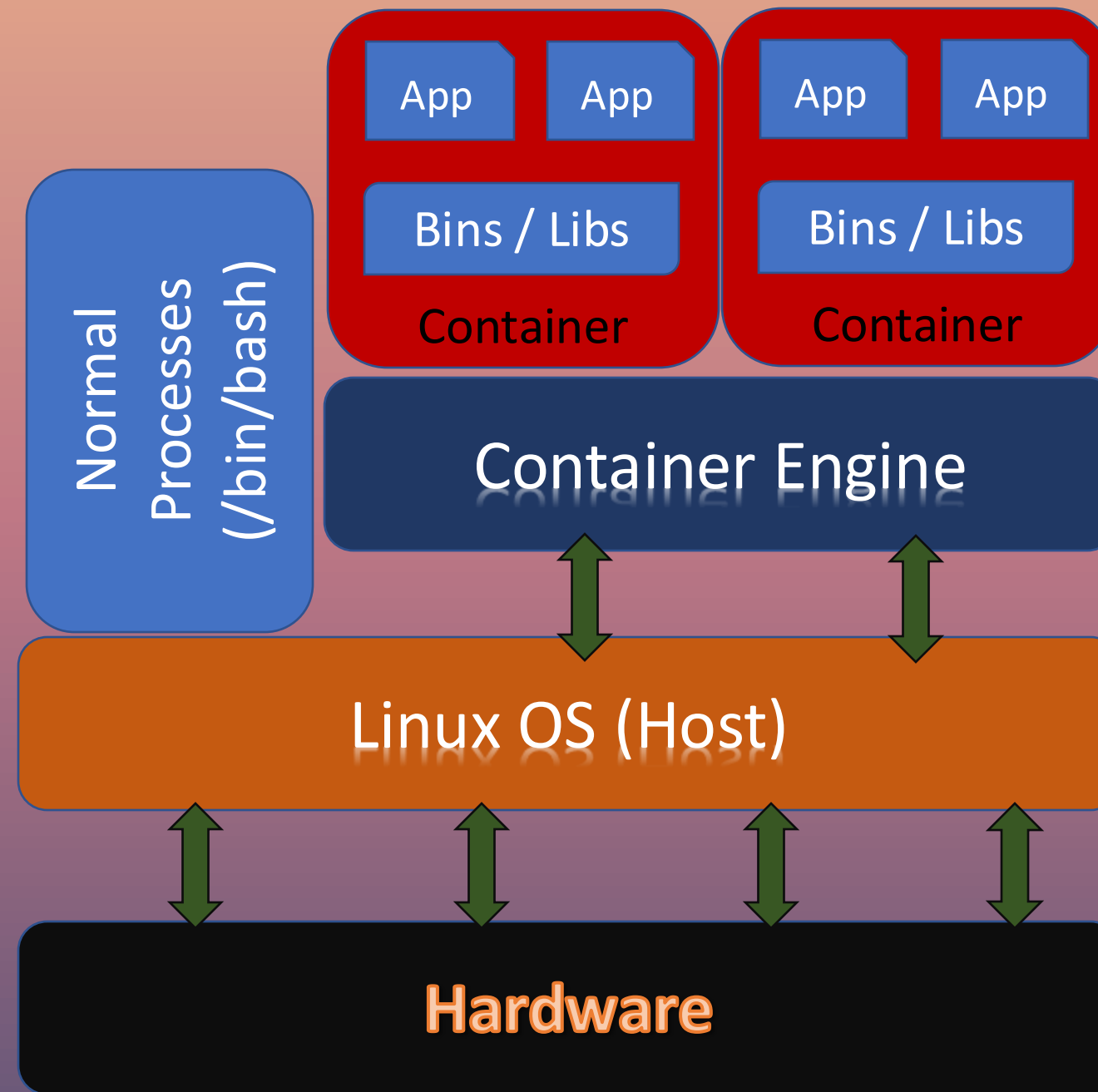
Introduction on LXC

Linux Container

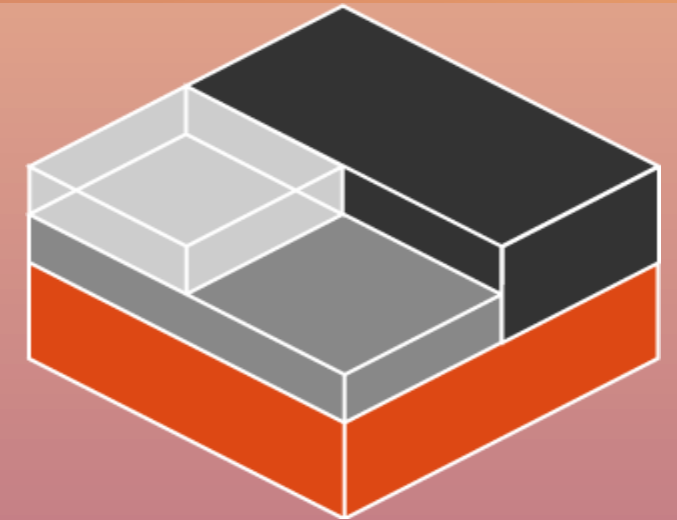
- ✓ Provides Operating System level virtualization
- ✓ Alternative to Hypervisors
- ✓ Running multiple, isolated Linux systems on a single Linux Control host
- ✓ Guests share the same kernel
- ✓ Guests perform like a standalone server
- ✓ Guests work as a Userspace interface
- ✓ No emulation, runs as a process on the host
- ✓ Limited to Linux guests only



Architecture of LXC

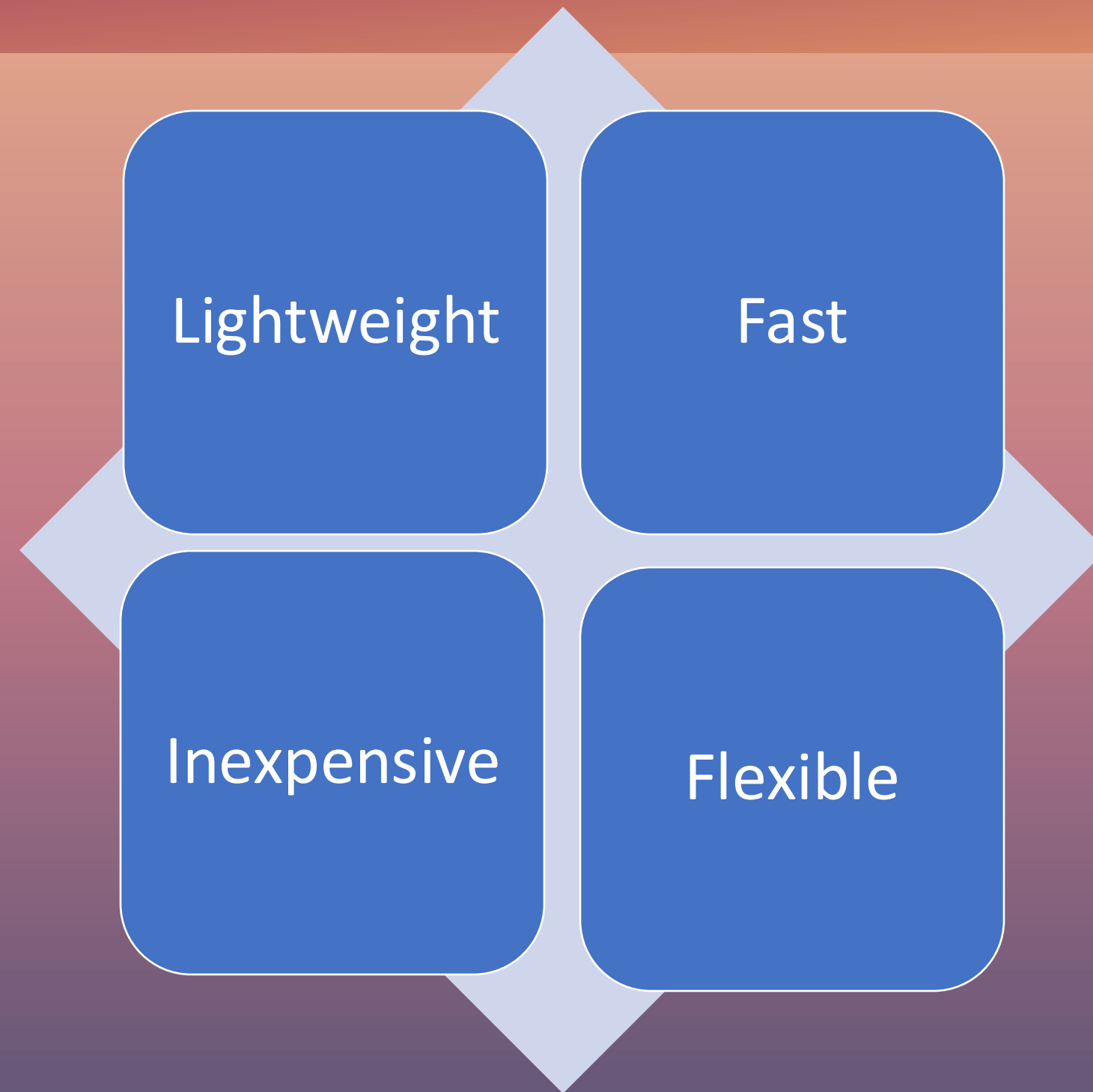


Linux Container



Why Linux Containers (LXC)

Linux Container



TTT Concept

Technology, Technique, Tool

TECHNOLOGY



KVM, LXC

TECHNIQUE

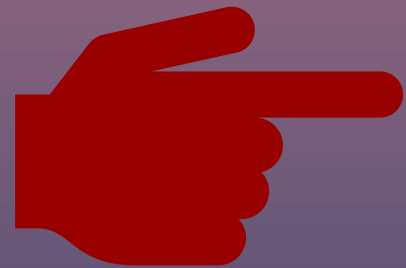


Virtualization
Containerization

TOOL



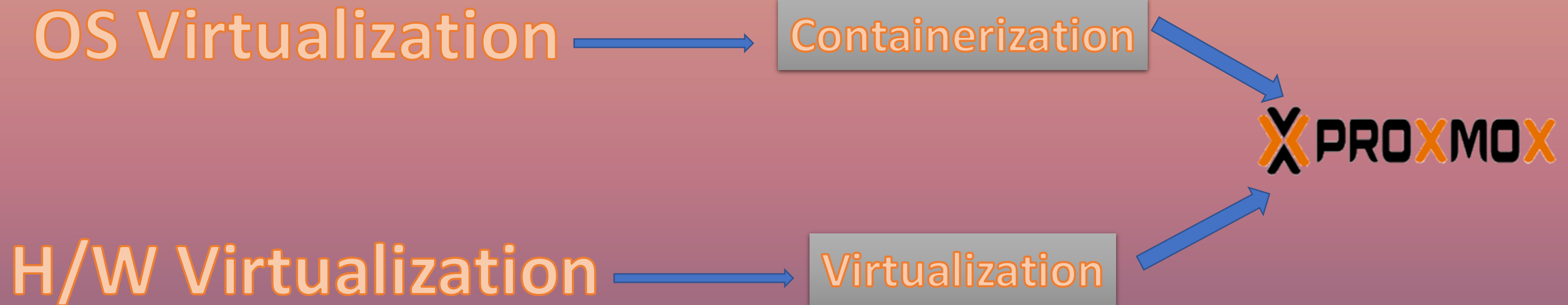
Proxmox VE



Proxmox internally uses KVM for full virtualization and LXC for containers.

Clarifying the Ambiguity

Virtualization, Containerization



XPROXMOX Stack

User Tools

qm

pct

pvesm

pvecm

pveum

pveceph

ha-manager

pve-firewall

Services

pveproxy

pvedaemon

pvestatd

pve-ha-lrm

pve-cluster

VM 101

App App

Guest OS

VM 100

App App

Guest OS

...

CT 200

App App

CT Distro

CT 201

App App

CT Distro

...

QEMU

KVM

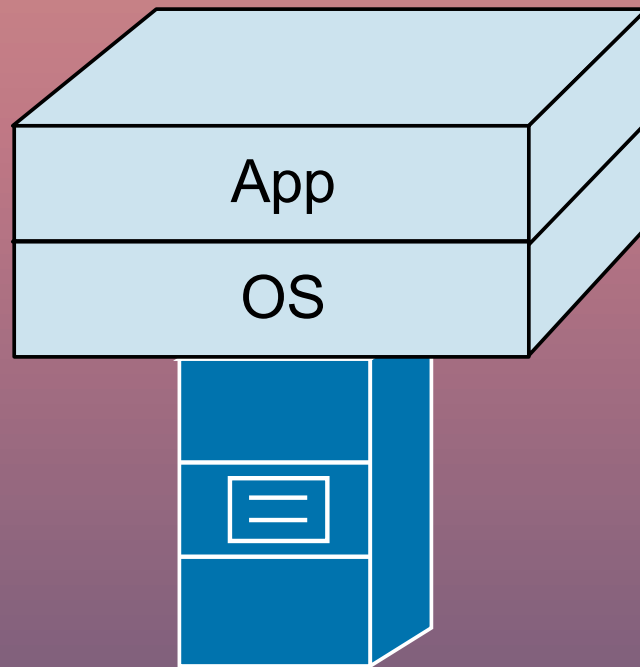
LXC

Linux Kernel (Debian)

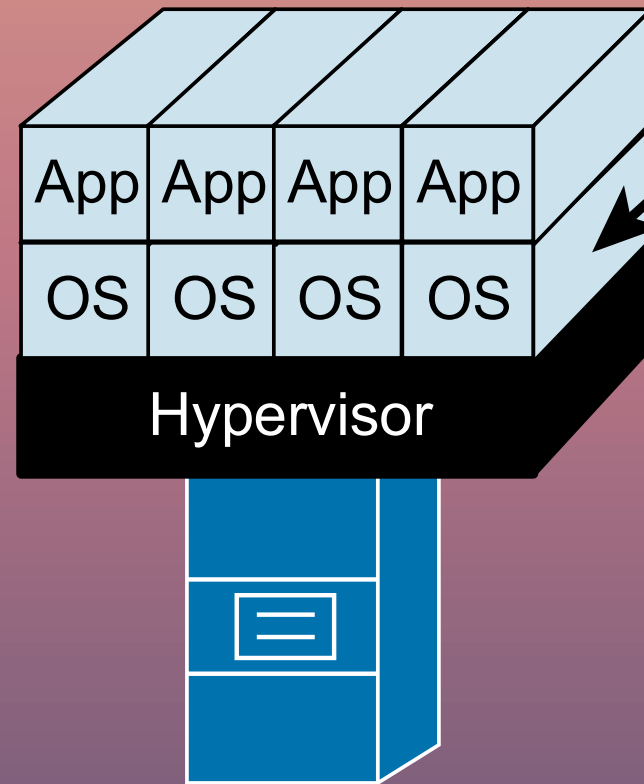
Normal
Processes
(/bin/bash)

Server Virtualization Technology

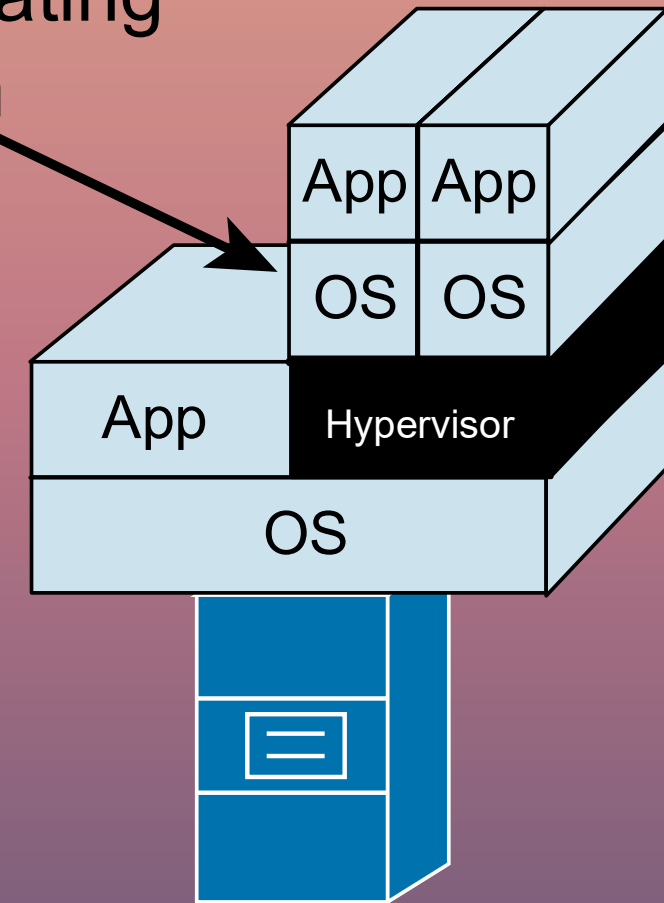
Physical Server



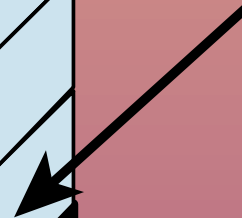
Type-1



Type-2



Guest Operating
System



XPROXMOX VE Features

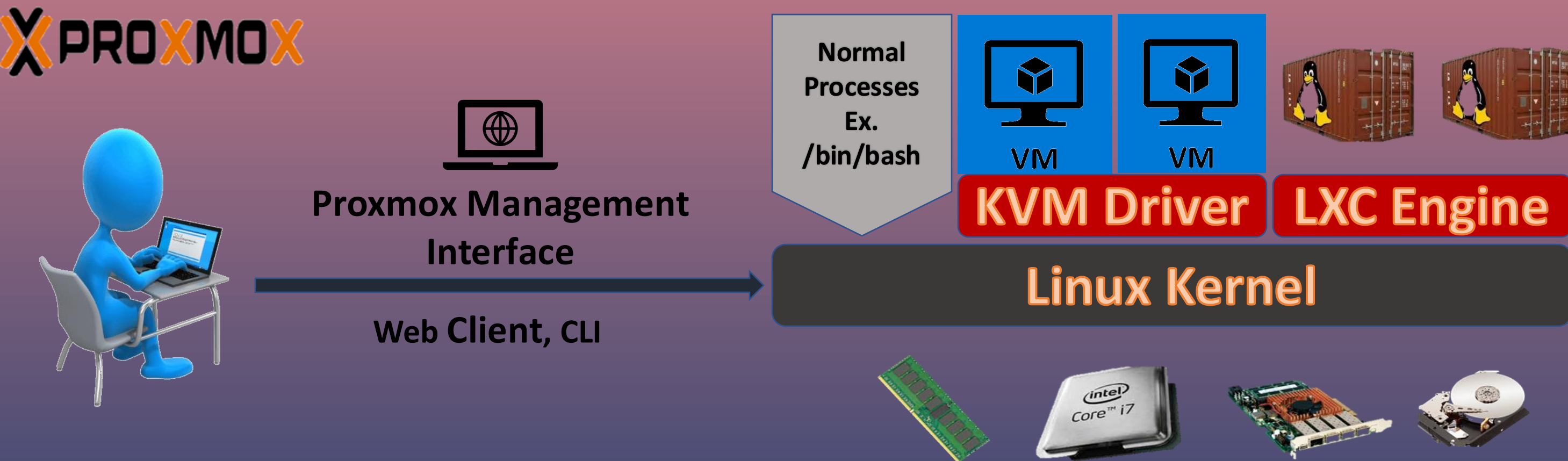
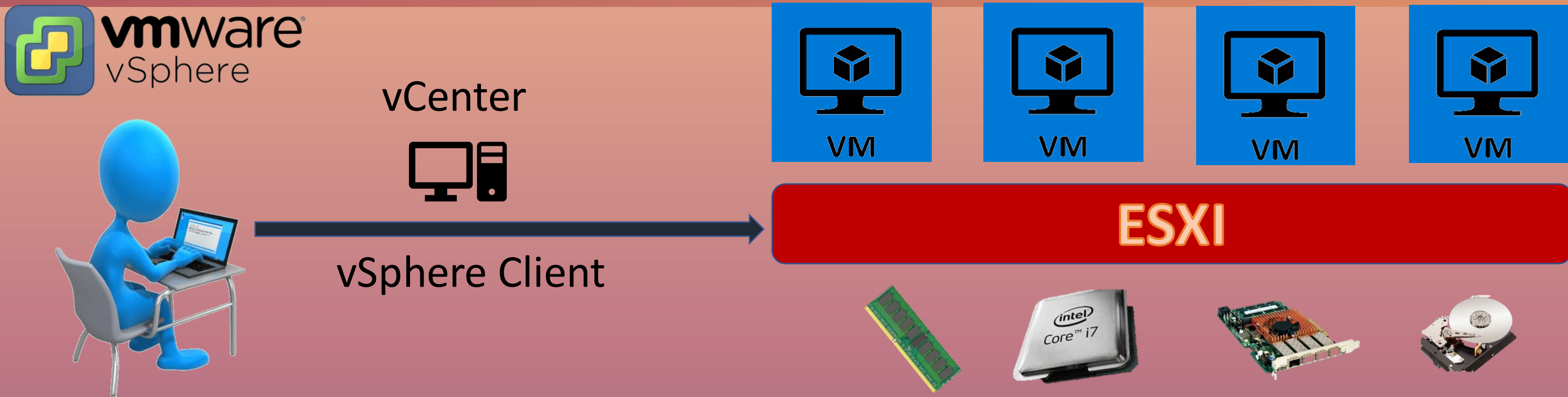
FEATURES

- 1 Built-in Web-based Multi-Master Management
- 2 Flexible Storage Model
- 3 Virtualization and Containerization
- 4 High Availability Cluster
- 5 Hyper-converged Infrastructure
- 6 Nested Virtual Environment
- 7 Integrated Backup and Restore



Hypervisor types

Linux Container

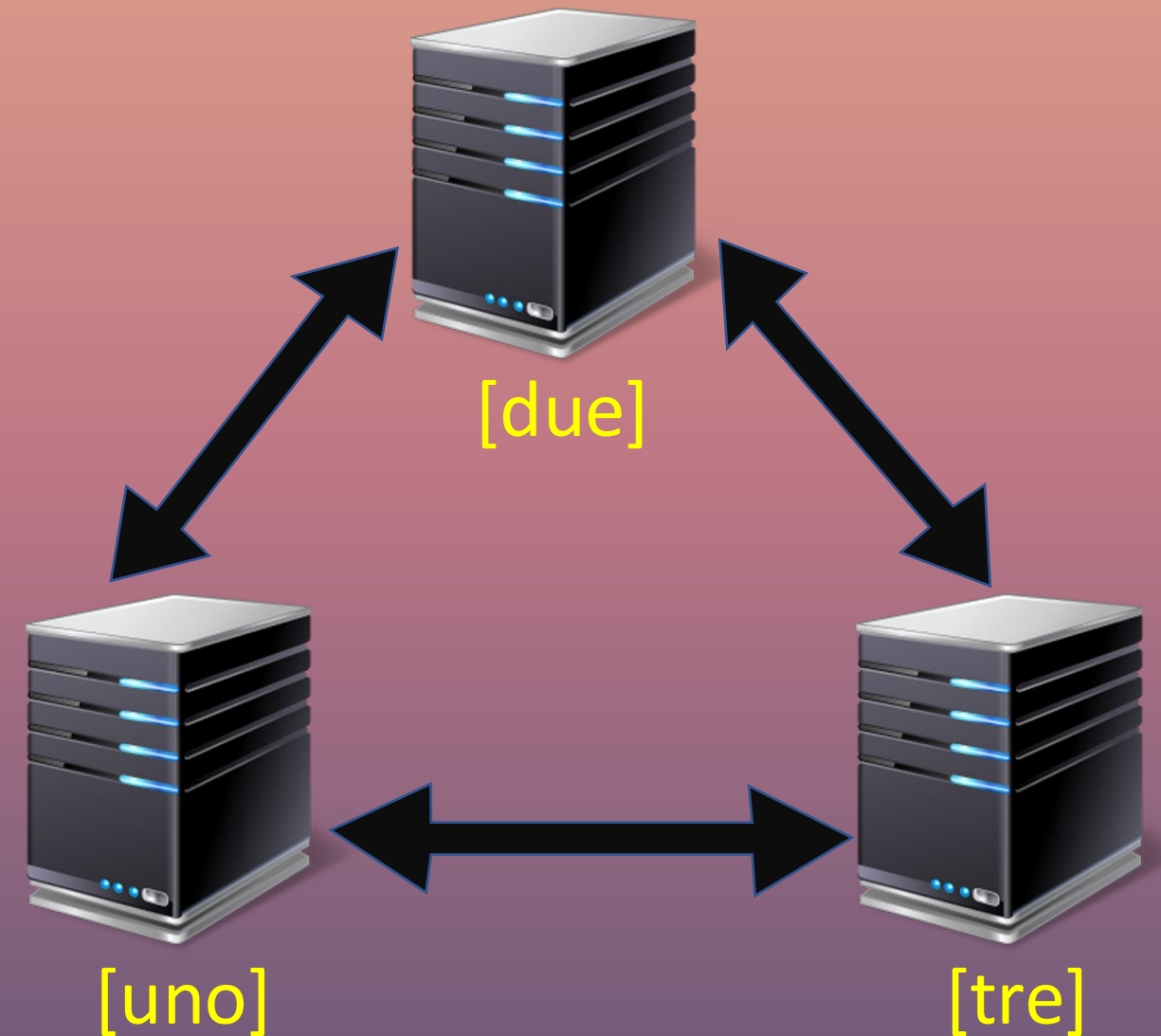


Proxmox VE Cluster

Clustering

GROUPING NODES INTO A CLUSTER HAS THE FOLLOWING ADVANTAGES:

- ✓ *Centralized, web based management*
- ✓ *Access the whole cluster on every node*
- ✓ *Unique multi-master design*
- ✓ *Easy migration of VM and CT between physical hosts*
- ✓ *Fast Deployment*
- ✓ *No single point of failure*



Proxmox VE Cluster

Clustering

- Proxmox VE cluster manager (pvecm) is a tool used to create a group of physical servers.
- Corosync Cluster Engine is a technology used for reliable group communication.
- There's no explicit limit for the number of nodes in a cluster.
- Proxmox Cluster File System (pmxcfs) is a database-driven file system for storing configuration files, which are replicated in real-time to all cluster nodes using Corosync.
- Because of using pmxcfs, you can connect to any node to manage the entire cluster. There is no need for a dedicated manager node.

Networking with Proxmox VE

Network Models

- 1 Bridged
- 2 Masquerading with NAT
- 3 Routed configuration
- 4 VLAN support
- 5 NIC Bonding



Comparison between VMware and proxmox

Features	Proxmox VE	VMware
Open Source	✓	✗
Integrated management Interface	Built-in Web management interface and shell-based CLI	Requires dedicated management server at additional cost
Supports Containers	✓	✓
Community Support	✓	✗
Host Isolation	Medium	Strong and Robust
Security	Less Secured	Highly Secured
Fault Tolerant	✓	✓
Live Migration	✓	✓
High Availability	✓	✓
Live VM Snapshots	✓	✓

Choosing a Network Configuration

PVE server in a private LAN, using an external gateway to reach the Internet

- **Bridged Model**

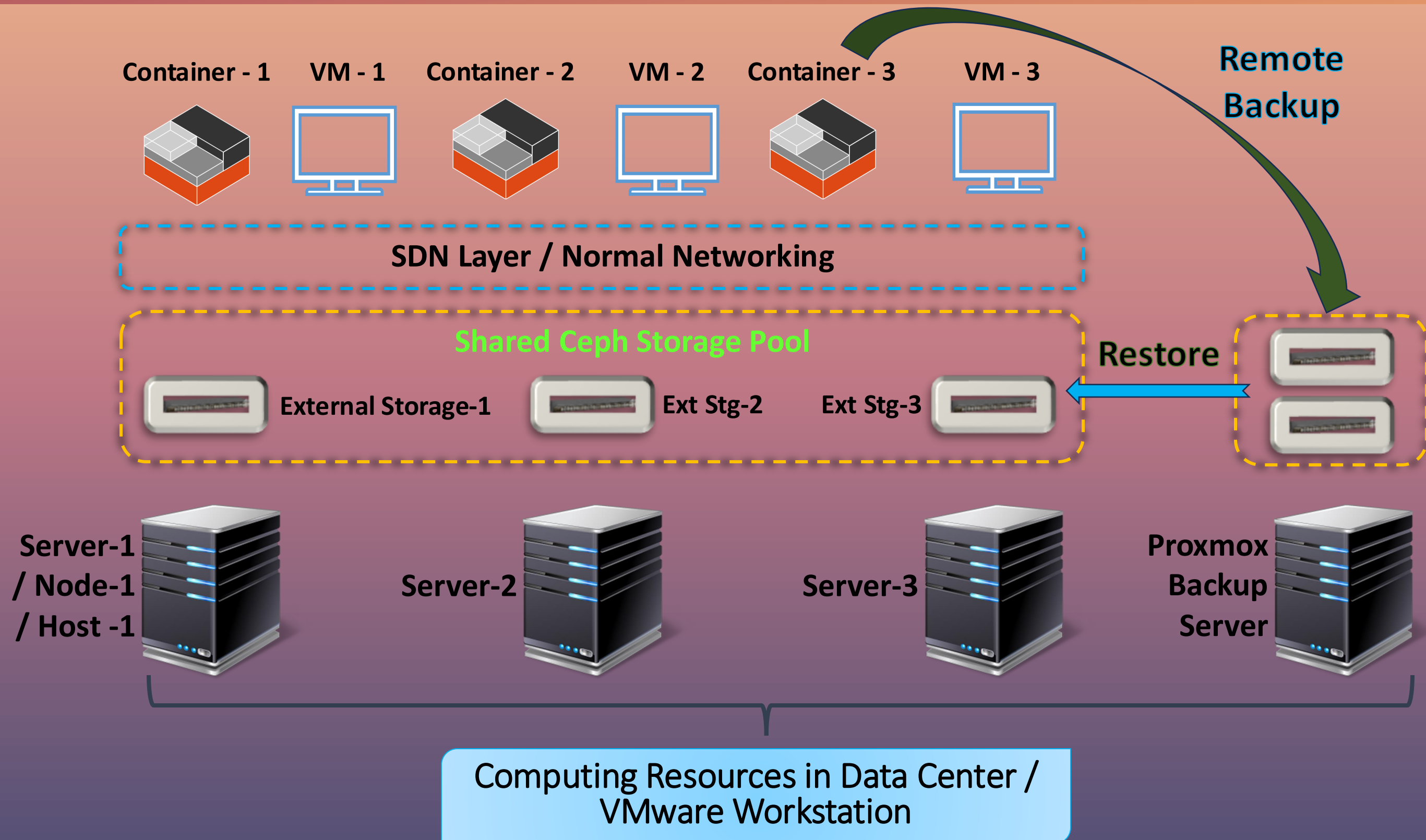
PVE server at hosting provider, with public IP ranges for guests

- **Bridged or Routed Model**

PVE server at hosting provider, with a single public IP address

- *Masquerading with NAT*

Lab Architecture for building Cloud Infrastructure



THANK YOU
Any Questions ?



Jamilur Rahman



+8801711377759



jamil@bdren.net.bd